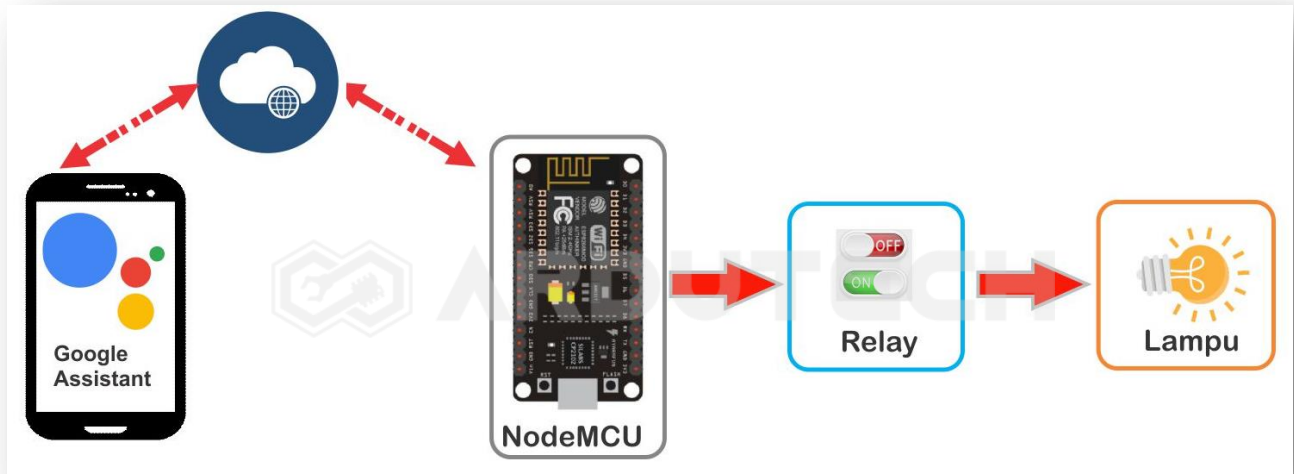


Kontrol 4 Relay dengan Google Assistant

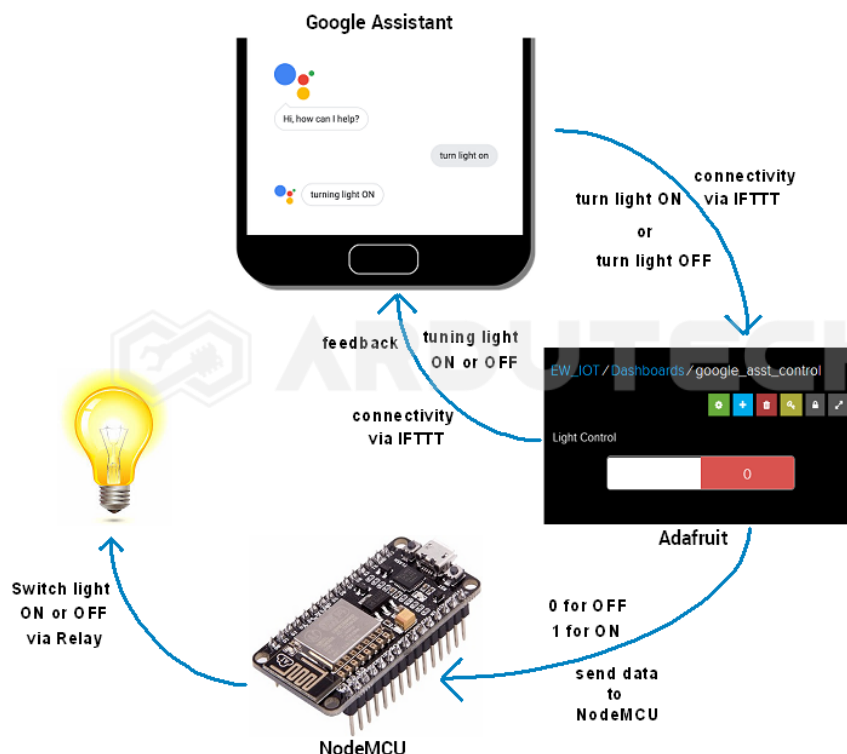
Deskripsi

Kontrol On – Off 4 relay melalui perintah suara “Google Assistant” di aplikasi Android. Kita cukup mengucapkan “1 ON” maka Relay 1 akan aktif, atau “Relay 1 off” dan relay 1 akan off. Dalam proyek ini ada 4 relay yang dapat dikontrol dan masing – masing relay dapat dihubungkan dengan peripheral rumah tangga seperti lampu AC, kipas angin dll.



Cara Kerja

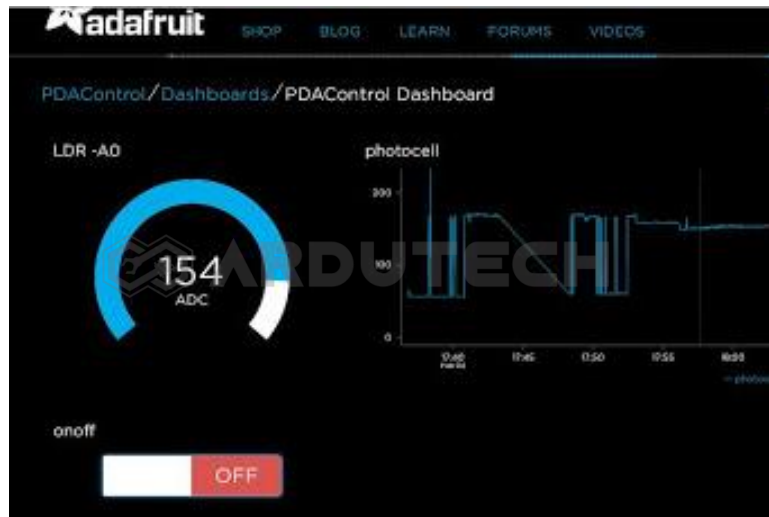
NodeMCU ESP8266 memanfaatkan layanan MQTT server yang disediakan oleh Adafruit IO akan memproses command yang diterima dari IFTTT yang terintegrasi dengan “Google Assistant” kemudian diterjemahkan menjadi sinyal kontrol ON – OFF untuk mengendalikan relay.



Pada proyek ini kita memerlukan beberapa aplikasi :

- **Adafruit IO**
- **IFTTT**
- **Google Assistant**

Adafruit IO (<https://io.adafruit.com/>) adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (*Internet of Things*) . Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



IFTTT (*If This Then That*) adalah sebuah layanan sistem otomasi yang menggabungkan beberapa layanan internet menjadi satu (<https://ifttt.com>). Service IFTTT secara garis besar digambarkan seperti ini, “Jika terjadi A, maka lakukan hal B”. Tidak hanya layanan web, IFTTT juga dapat digunakan untuk mengotomasi peralatan yang terhubung dengan internet seperti termostat atau lampu pintar.

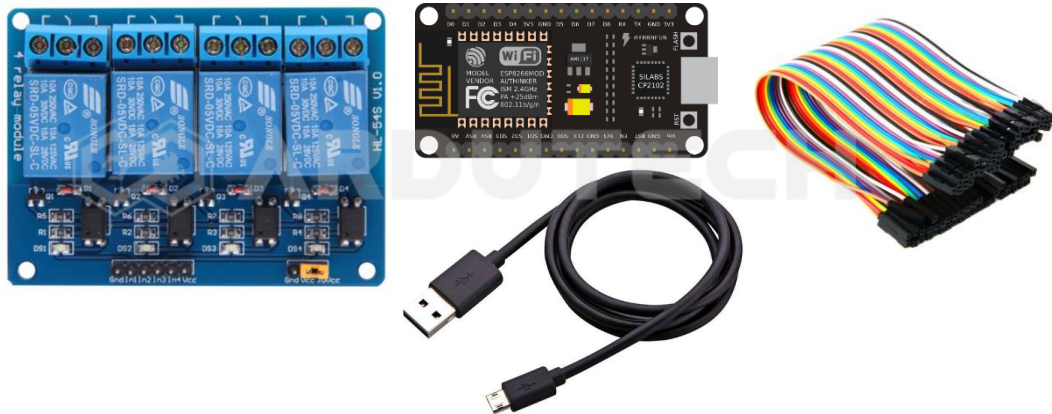
Secara sederhana, setiap aktivitas yang kamu lakukan secara online dapat diatur untuk memiliki reaksi otomatis pada sebuah aplikasi web tertentu.

if  this then that

 Want to build your own service? Build on the platform 

Kebutuhan Bahan

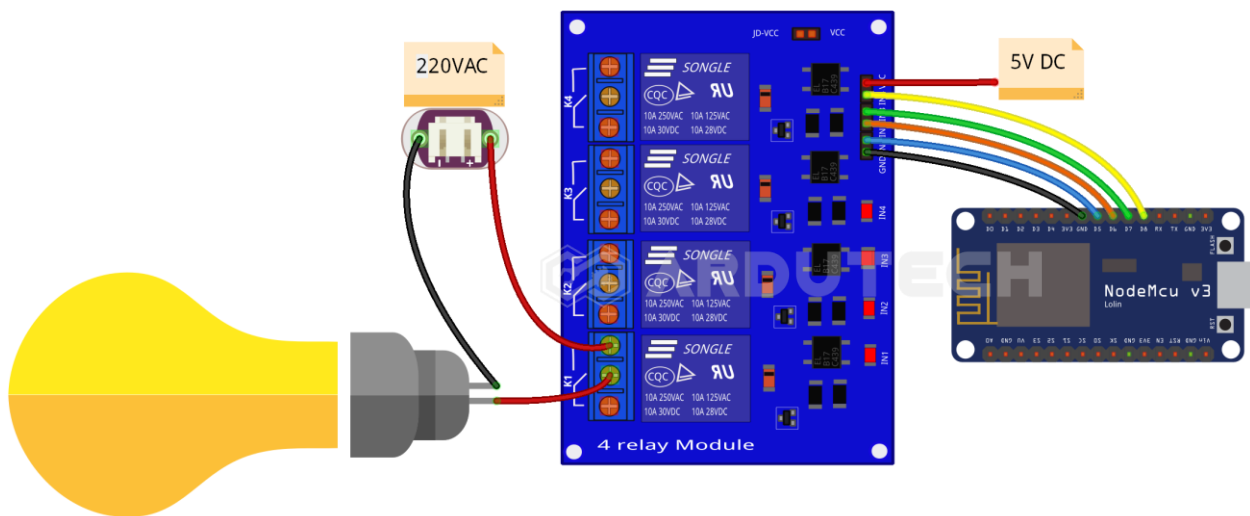
- NodeMCU ESP8266 V3
- Modul relay 4 channel 5V
- Kabel micro USB
- Kabel konektor



Kebutuhan Software

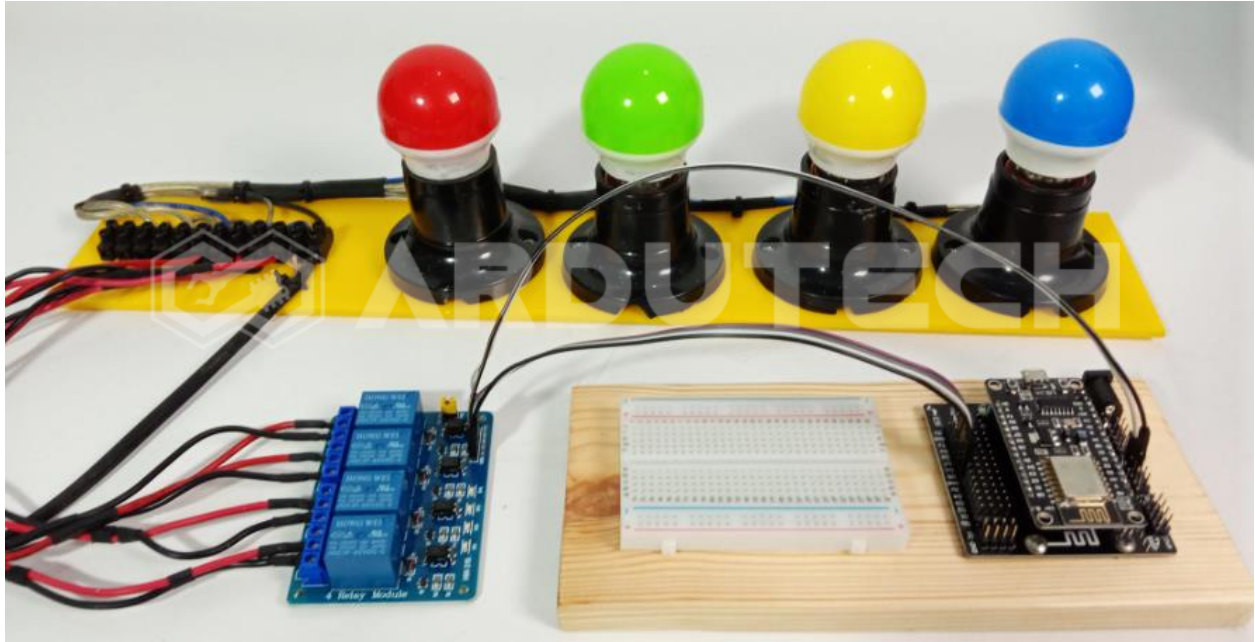
- Arduino IDE

Rangkaian/Skematik



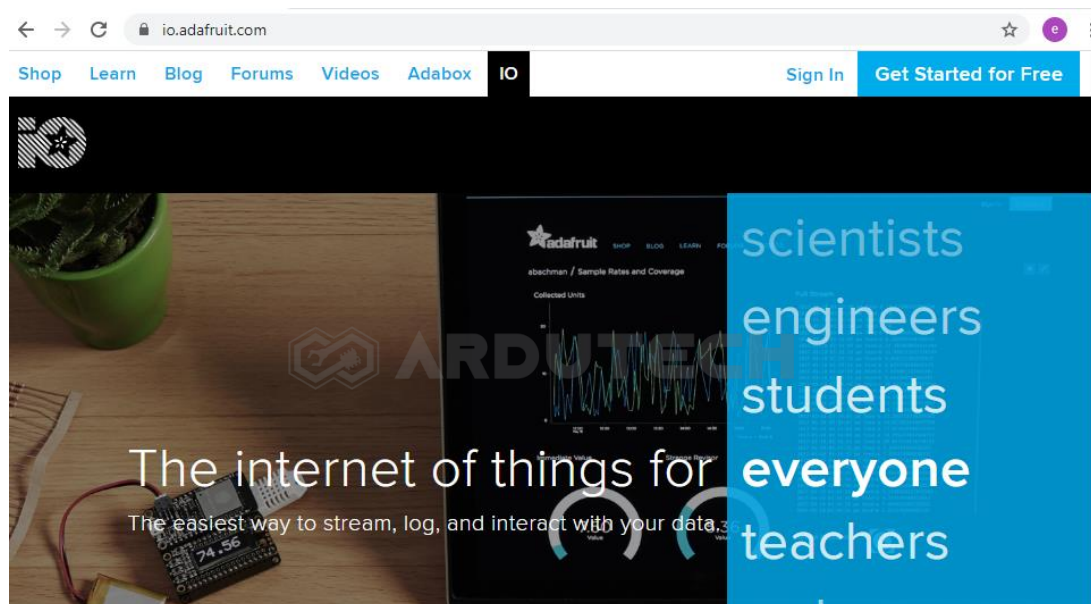
Pada rangkaian hanya ada 1 lampu AC, untuk 3 perangkat yang lain anda tinggal menambahkan sama dengan lampu 1.

NodeMCU V3	4 Ch Relay Module
D5	IN1
D6	IN2
D7	IN3
D8	IN4



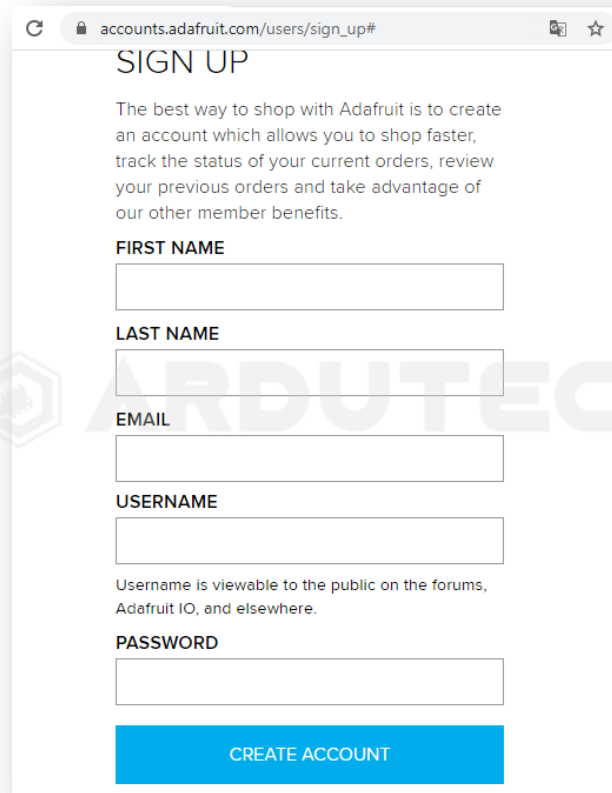
Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>



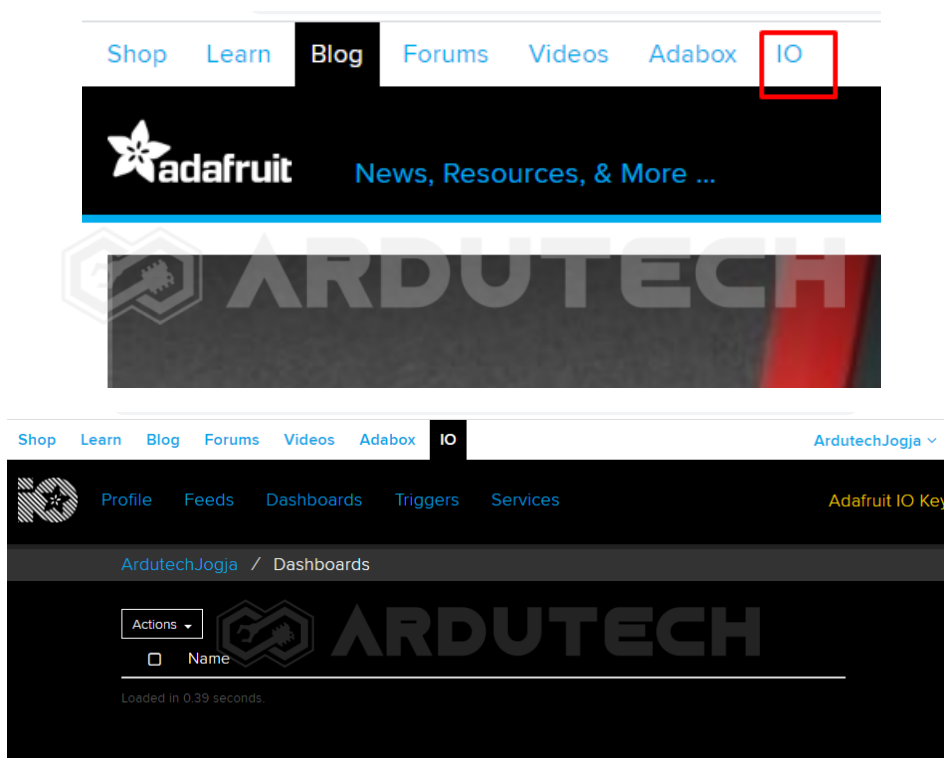
Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

Klik pada **“Get Started for free”** di bagian pojok kanan atas.

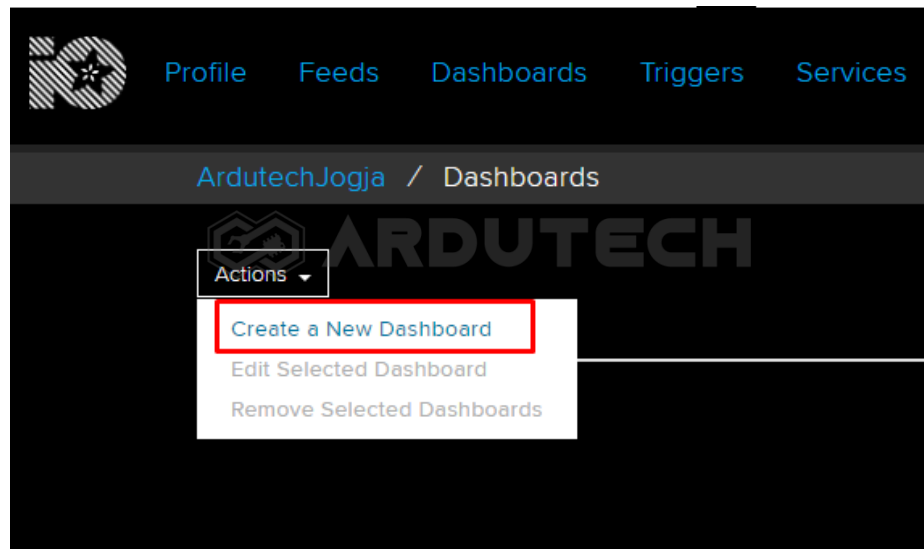


Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

Masuk ke menu **IO**.

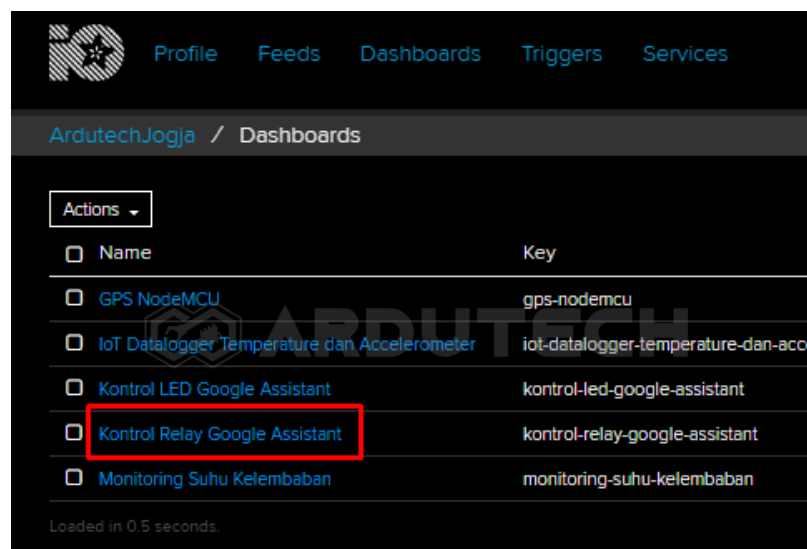


Klik pada tombol **“Actions”** kemudian pilih **“Create a New Dashboard”**

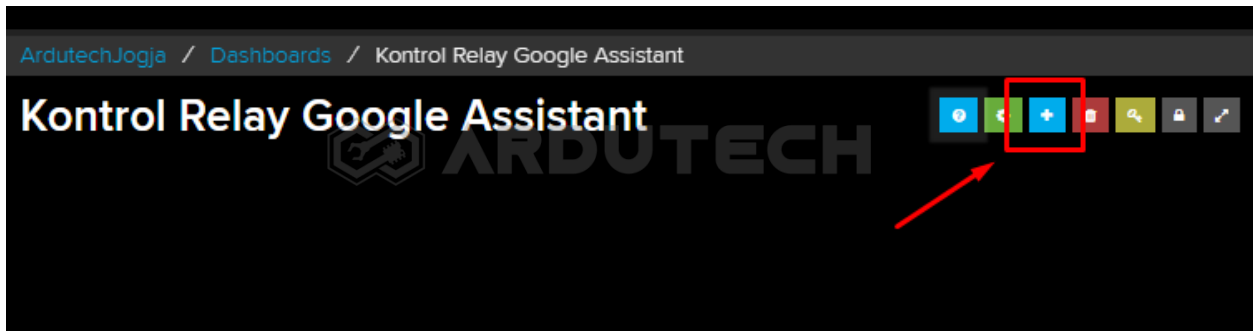


Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya **“Kontrol Relay Google Assistant”** dan klik **“Create”**.

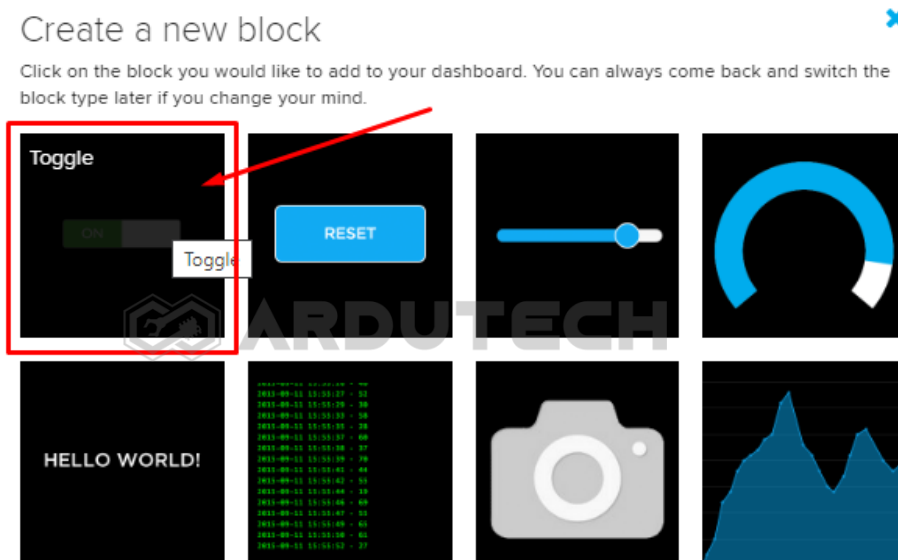
Setelah klik tombol **“Create”**, akan muncul dashboard baru dengan nama **“Kontrol Relay Google Assistant”**.



Klik “Kontrol Relay Google Assistant” kemudian pilih tombol “+” untuk membuat blok baru.



Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Toggle** (pojok kiri atas).



Selanjutnya muncul tampilan untuk memilih ‘Choose feed’ . Ini semacam “Topik” dalam komunikasi MQTT. Jadi supaya dikenali oleh ‘subscriber’ (penerima perintah) maka ‘publisher’ harus menentukan ‘topik’ nya. Silakan tulis “Relay1” kemudian klik tombol “Create”.

A screenshot of a 'Create a new Feed' dialog box. The title is 'Create a new Feed' with a close button (X) in the top right. It contains two input fields: 'Name' and 'Description'. The 'Name' field has the text 'Relay1' entered. Below the 'Name' field, it says 'Maximum length: 128 characters. Used: 6'. The 'Description' field is empty. At the bottom right, there are two buttons: 'Cancel' and 'Create'.

Setelah tombol “Create” diklik muncul tampilan :

Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Enter new feed name

Create

Group / Feed	Last value	Recorded
☐ GPS	🔒	2 days
☐ gpslat	🔒 0.000000	2 days
☐ gpslng	🔒 0.000000	2 days
☐ GSMloc	🔒	2 days
☐ LED	🔒 0	about 1 hour
☒ Relay1	🔒	3 minutes
☐ Relay2	🔒	3 minutes
☐ Relay3	🔒	3 minutes
☐ Relay4	🔒	2 minutes

← Previous step

Next step >

Pilih **Relay1** kemudian klik “**Next step >**” untuk seting Block :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the “Create Block” button to send it to your dashboard.

Block Title (optional)

Relay 1

Button On Text

1

Button Off Text

0

Block Preview

Relay 1

0

Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

45

Published Value

0 bytes

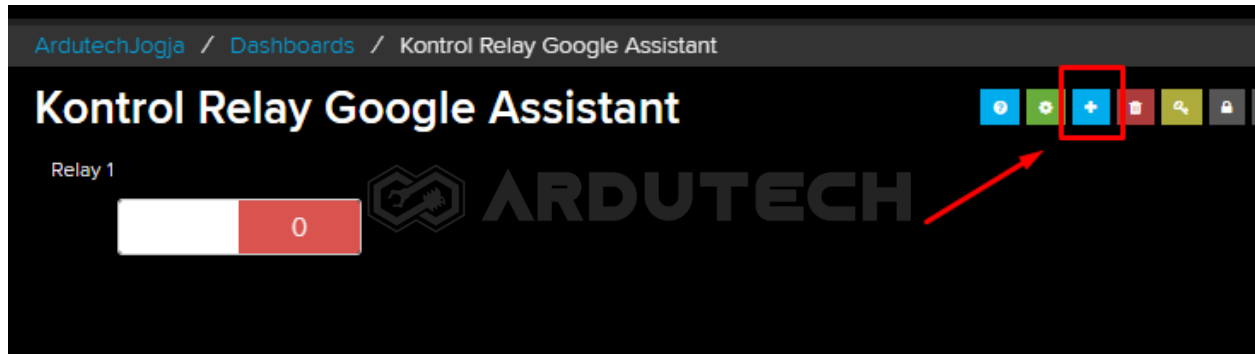
← Previous step

Create block

Isi pada kolom yang tersedia :

- Block Title : Relay 1 (opsional)
- Button On Text : 1
- Button Off Text : 0

Klik tombol “**Create block**”



Tambahkan **Block** lagi, klik tanda plus “+” kemudian pilih “**Toggle**” yang ada di pojok kiri atas.



Selanjutnya muncul tampilan untuk memilih ‘**Choose feed**’ . Silakan tulis “**Relay2**” kemudian klik tombol “**Create**”.

Create a new Feed

Name

Relay2

Maximum length: 128 characters. Used: 6

Description

Cancel

Create

Setelah tombol **“Create”** diklik muncul tampilan :

Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☐ GPS		2 days
☐ gpslat	0.000000	2 days
☐ gpslng	0.000000	2 days
☐ GSMloc		2 days
☐ LED	0	about 1 hour
☐ Relay1		5 minutes
☒ Relay2		5 minutes
☐ Relay3		4 minutes
☐ Relay4		4 minutes

Previous step

Next step

Pilih **Relay2** kemudian klik **“Next step >”** untuk seting Block :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Button On Text

Button Off Text

Block Preview

Relay 2



Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

Published Value

0 bytes

← Previous step

Create block

Isi pada kolom yang tersedia :

- Block Title : Relay 2 (opsional)
- Button On Text : 1
- Button Off Text : 0

Klik tombol **"Create block"**.

Tambahkan **Block** lagi, klik tanda plus "+" kemudian pilih **"Toggle"** yang ada di pojok kiri atas.

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Choose feed' . Silakan tulis **"Relay3"** kemudian klik tombol **"Create"**.

Create a new Feed

Name

Relay3

Maximum length: 128 characters. Used: 6

Description

Cancel Create

Setelah tombol **"Create"** diklik muncul tampilan :

Choose feed ✕

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☐ GPS	🔒	2 days
☐ gpslat	🔒 0.000000	2 days
☐ gpslng	🔒 0.000000	2 days
☐ GSMloc	🔒	2 days
☐ LED	🔒 0	about 1 hour
☐ Relay1	🔒	6 minutes
☐ Relay2	🔒	6 minutes
☒ Relay3	🔒	5 minutes
☐ Relay4	🔒	5 minutes

◀ Previous step
Next step ▶

Pilih **Relay3** kemudian klik **"Next step >"** untuk seting Block :

Block settings



In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Button On Text

Button Off Text

Block Preview



Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

Published Value



0 bytes

< Previous step

Create block

Isi pada kolom yang tersedia :

- Block Title : Relay 3 (opsional)
- Button On Text : 1
- Button Off Text : 0

Klik tombol **"Create block"**.

Tambahkan **Block** lagi, klik tanda plus "+" kemudian pilih **"Toggle"** yang ada di pojok kiri atas.

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Choose feed' . Silakan tulis "Relay4" kemudian klik tombol "Create".

Create a new Feed

Name

Relay4

Maximum length: 128 characters. Used: 6

Description

Cancel Create

Setelah tombol "Create" diklik muncul tampilan :

Choose feed



Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☐ GPS		2 days
☐ gpslat	0.000000	2 days
☐ gpslng	0.000000	2 days
☐ GSMloc		2 days
☐ LED	0	about 1 hour
☐ Relay1		7 minutes
☐ Relay2		7 minutes
☐ Relay3		6 minutes
☒ Relay4		6 minutes

< Previous step

Next step >

Pilih **Relay4** kemudian klik “**Next step >**” untuk seting Block :

Block settings



In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Button On Text

Button Off Text

Block Preview

Relay 4



Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

Published Value



0 bytes

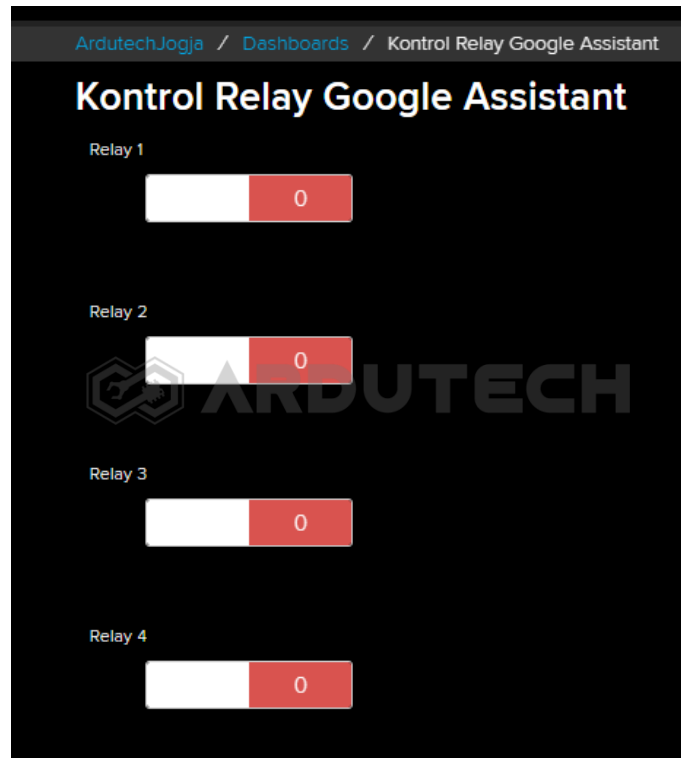
◀ Previous step

Create block

Isi pada kolom yang tersedia :

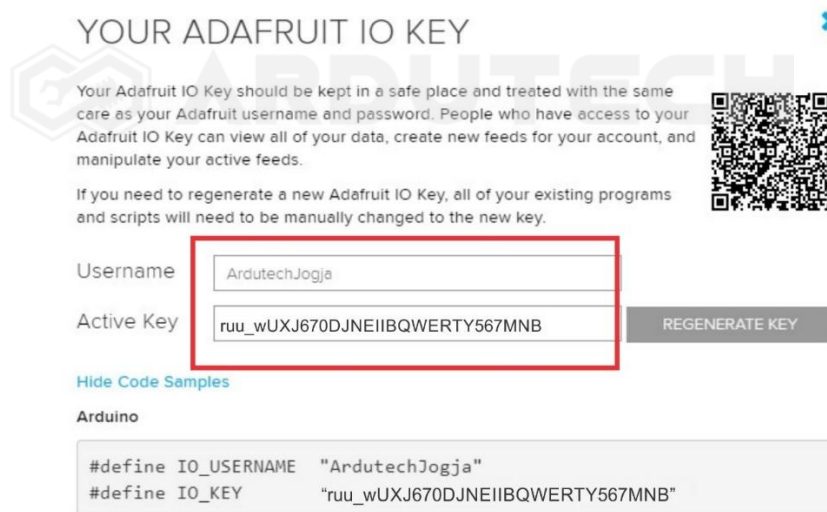
- Block Title : Relay 4 (opsional)
- Button On Text : 1
- Button Off Text : 0

Klik tombol **"Create block"**.



Ok sekarang sudah ada 4 block dengan masing – masing toggle (saklar).

Klik bagian “**Adafruit IO Key**” di pojok kanan atas.



Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut.

Nah untuk pembuatan antarmuka pada Adafruit sudah selesai, sekarang kita siapkan program Arduino IDE-nya.

Membuat Applet di IFTTT

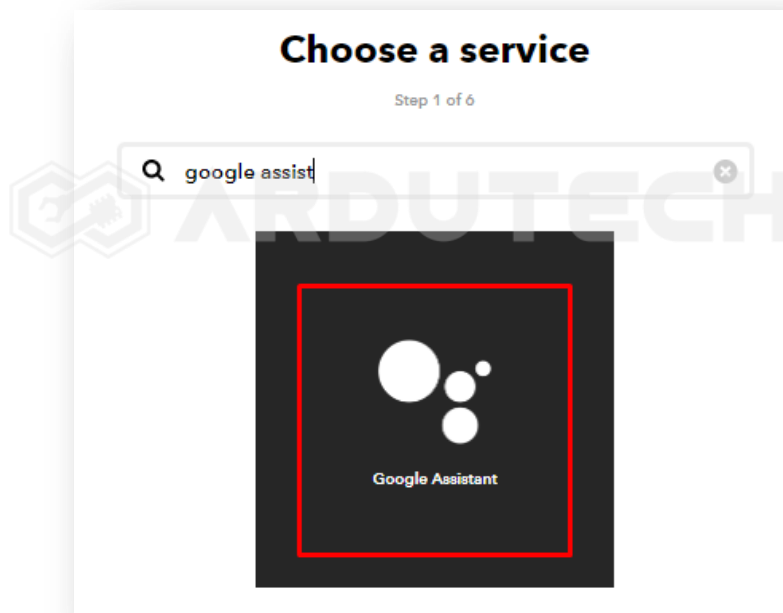
Buka halaman IFTTT di <https://ifttt.com/create>.

Pastikan anda telah mempunyai akun di IFTTT. Jika belum punya akun silakan daftar terlebih dahulu. Gunakan akun google yang sama dengan akun Adafruit dan akun google di Android anda. Ok setelah tampilan pertama muncul :

Create your own

If  This Then That

Klik + This sehingga tampil :



Pada kolom pencarian ketik “**google assistant**”, setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back

Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for? [Suggest a new trigger](#)

Pilih "Say a simple phrase".

Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 1 on

What's another way to say it? (optional)

Relay one on

And another way? (optional)

One on

What do you want the Assistant to say in response?

Ok boss

Language

English

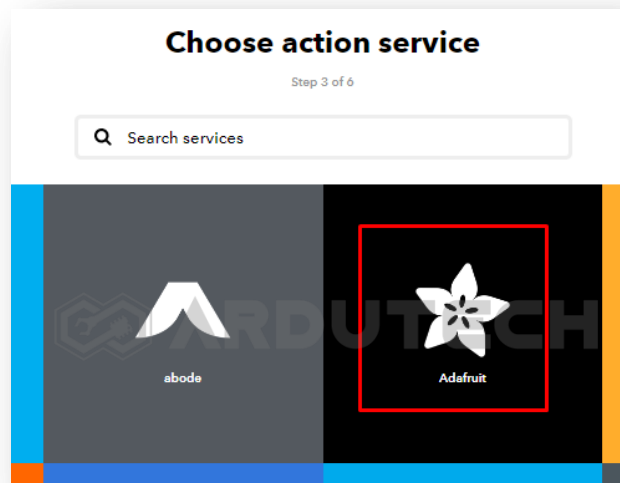
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik **"Create trigger"**.

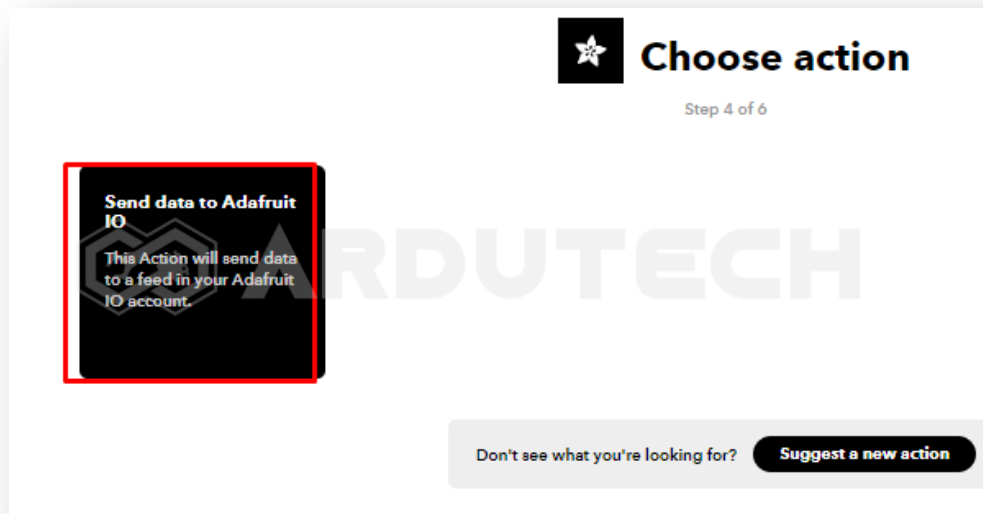
Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If Then That

Cari aplikasi /service dari Adafruit.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO.**



Complete action fields

Step 5 of 6

Feed name

Relay1

The name of the feed to save data to.

Data to save

1

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay1**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 1, maksudnya logika '1' untuk 'ON'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 1 on", then Send data to Relay1 feed

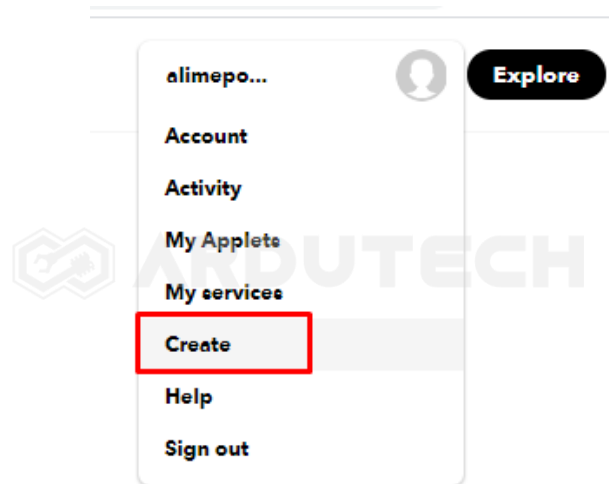
54/140

by alimepol22

Receive notifications when this Applet runs

Finish

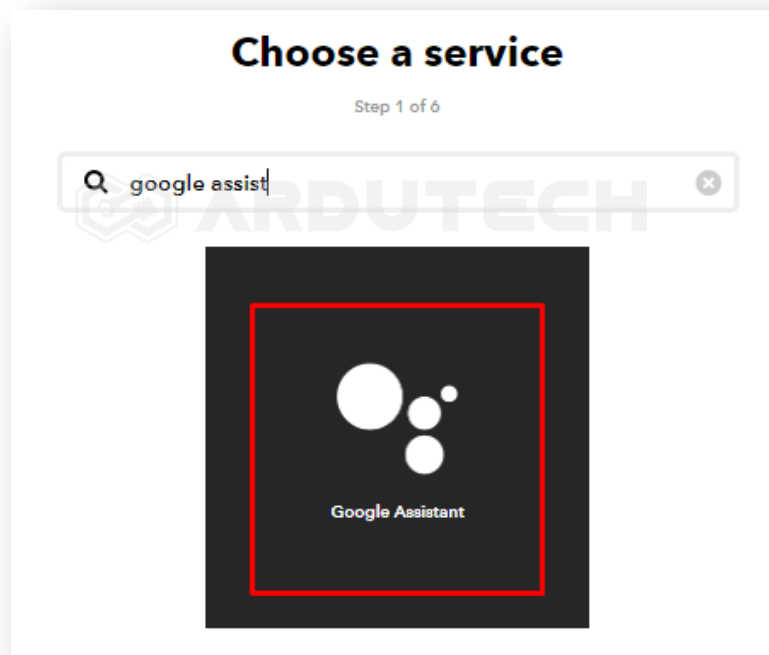
Terakhir klik **Finish**. Ok itu tadi baru bagian untuk perintah untuk Relay 1 ON, sekarang kita buat perintah Applet untuk mematikan relay (Relay 1 OFF). Pilih menu "**Create**" :



Tampilan dan proses sama dengan membuat Applet untuk mengaktifkan Relay 1.



Klik + **This** sehingga tampil :



Pada kolom pencarian ketik “**google assistant**”, setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".



Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 1 off

What's another way to say it? (optional)

Relay one off

And another way? (optional)

One off

What do you want the Assistant to say in response?

OK boss

Language

English



Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik **"Create trigger"**.

Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

www.ardutech.com @2020 (Terimakasih anda tidak meng-copy file ini untuk dijual kembali)

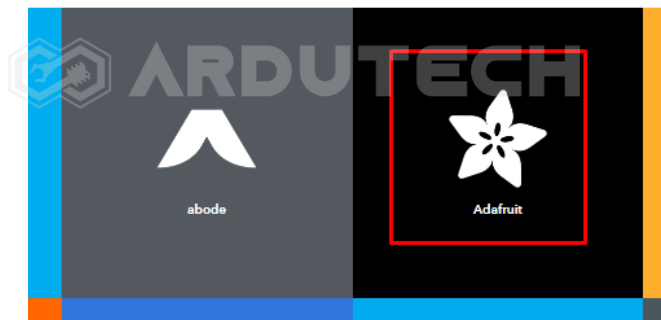
If Then That

Cari aplikasi /service dari **Adafruit**.

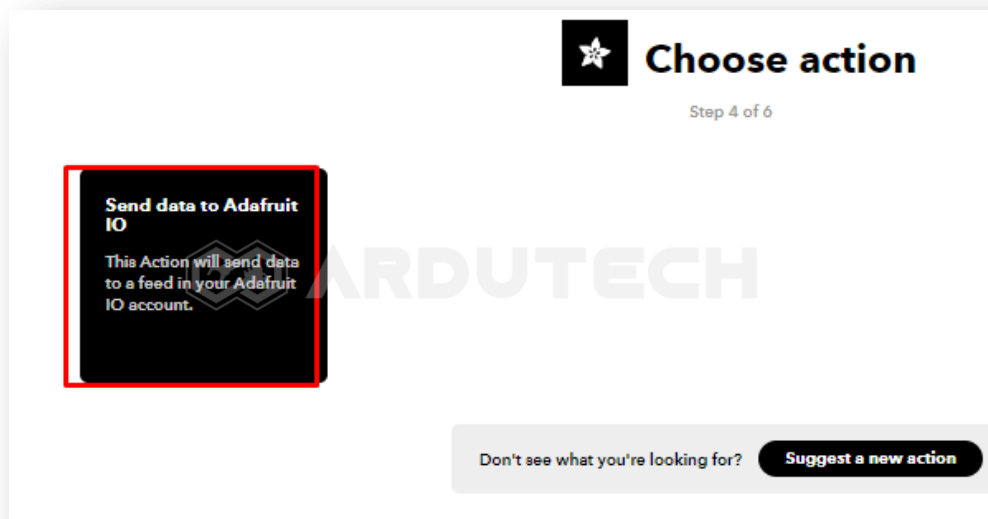
Choose action service

Step 3 of 6

Search services



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO.**



Complete action fields

Step 5 of 6

Feed name

Relay1

The name of the feed to save data to.

Data to save

0

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay1**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 1 off",
then Send data to
Relay1 feed

55/140

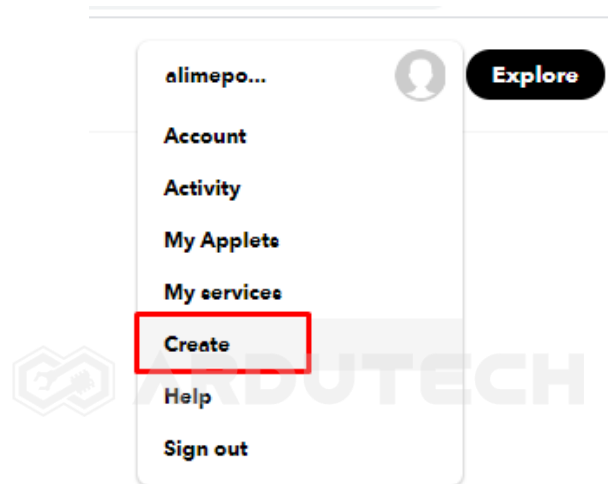
by alimepol22

Receive notifications
when this Applet runs

Finish

Terakhir klik **Finish**.

Kita lanjutkan untuk membuat Applet Relay 2 yang terdiri dari 2 perintah : Relay 2 ON dan Relay 2 OFF. Pilih menu "**Create**" :

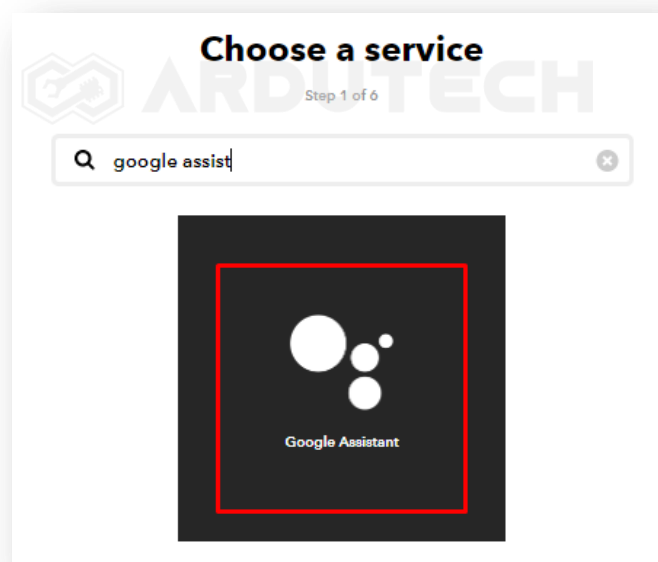


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.

Create your own

If **+** This Then That

Klik + This sehingga tampil :



Pada kolom pencarian ketik “google assistant” , setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".



Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 2 on

What's another way to say it? (optional)

Relay two on

And another way? (optional)

Two On

What do you want the Assistant to say in response?

Ok boss

Language

English



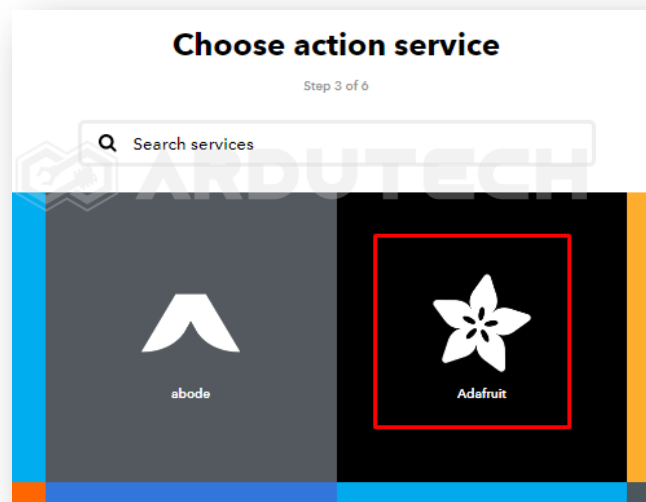
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik **"Create trigger"**.

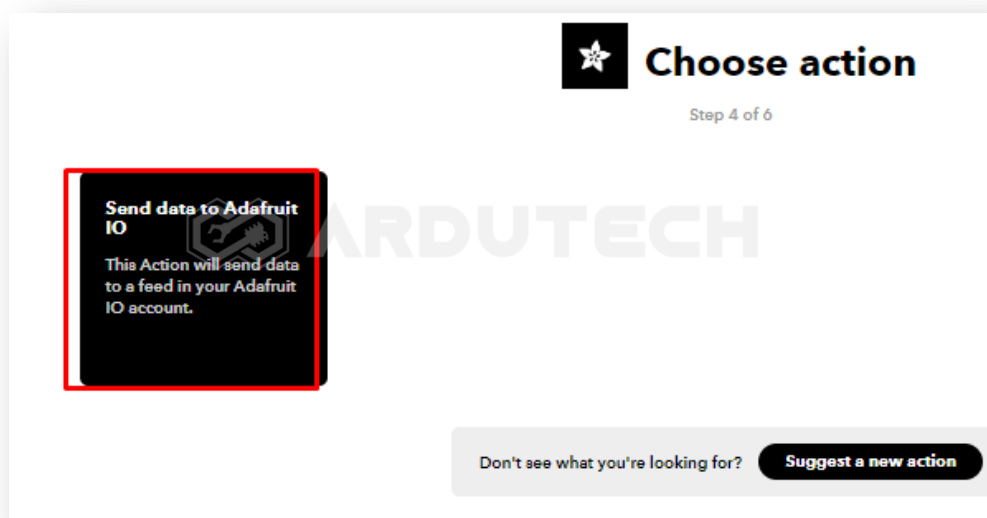
Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.



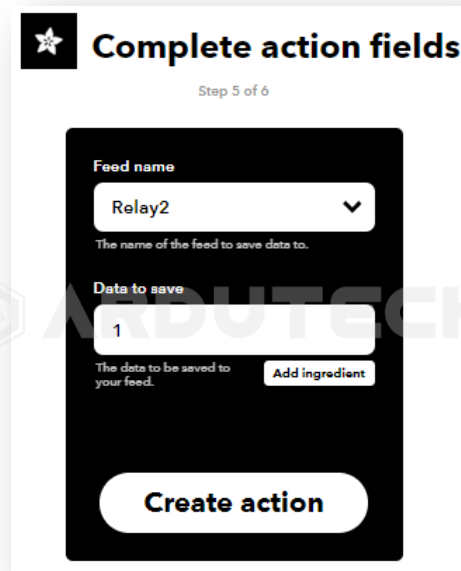
Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields
Step 5 of 6

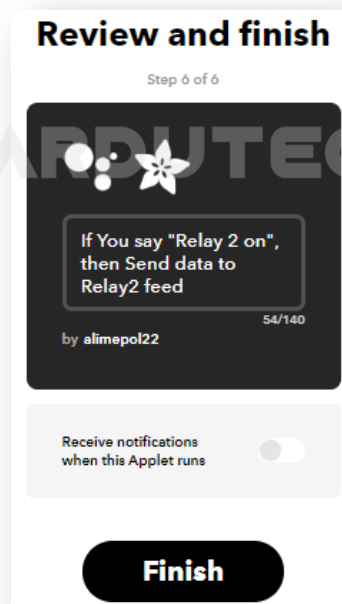
Feed name
Relay2
The name of the feed to save data to.

Data to save
1
The data to be saved to your feed. [Add ingredient](#)

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay2**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.



Review and finish
Step 6 of 6

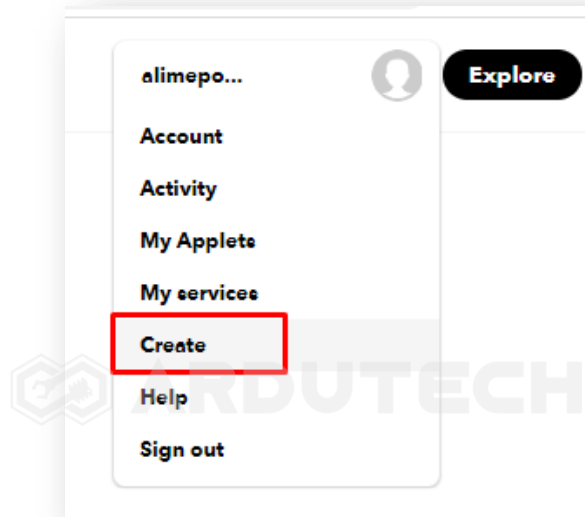
If You say "Relay 2 on",
then Send data to
Relay2 feed
54/140
by alimepol22

Receive notifications
when this Applet runs

Finish

Terakhir klik **Finish**.

Pilih lagi menu **"Create"** :

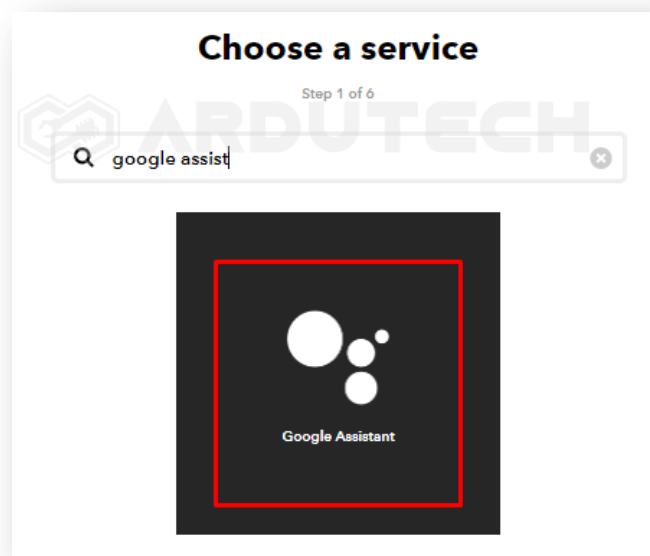


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.

Create your own

If + This Then That

Klik + **This** sehingga tampil :



Pada kolom pencarian ketik “**google assistant**” , setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".

Complete trigger fields

Step 2 of 6

What do you want to say?

What's another way to say it? (optional)

And another way? (optional)

What do you want the Assistant to say in response?

Language

English ▼

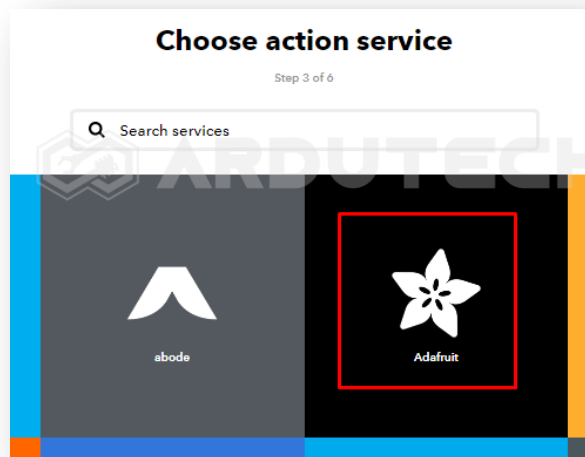
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik **"Create trigger"**.

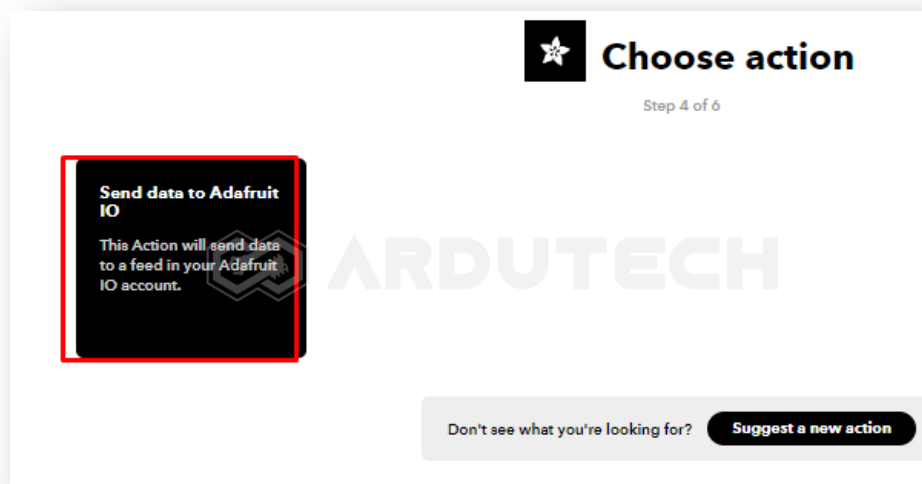
Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If  **Then**  **That**

Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields

Step 5 of 6

Feed name

Relay2

The name of the feed to save data to.

Data to save

0

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay2**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 2 Off", then Send data to Relay2 feed

55/140

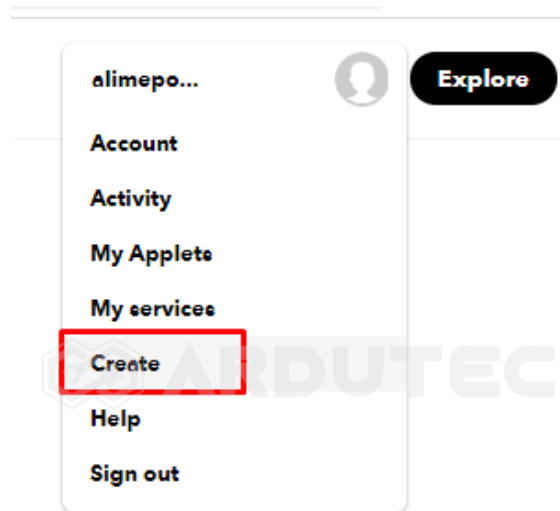
by alimepol22

Receive notifications when this Applet runs

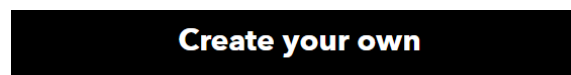
Finish

Terakhir klik **Finish**.

Pilih lagi menu **"Create"** :

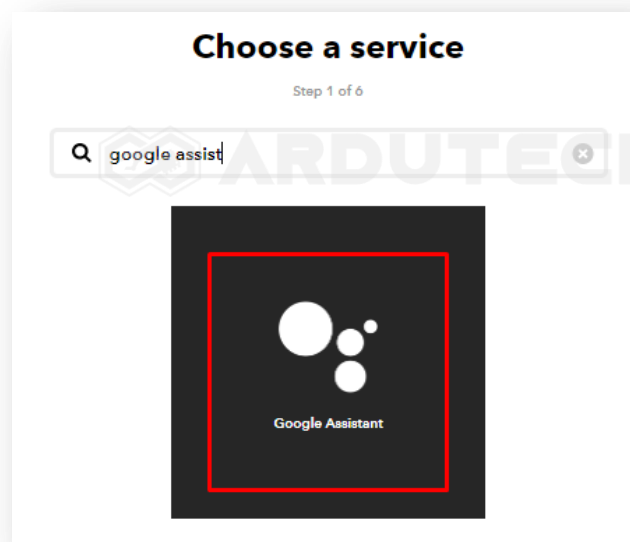


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.



If **+** This Then That

Klik + This sehingga tampil :



Pada kolom pencarian ketik “**google assistant**”, setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".



Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 3 On

What's another way to say it? (optional)

Relay three On

And another way? (optional)

Three On

What do you want the Assistant to say in response?

Ok boss

Language

English



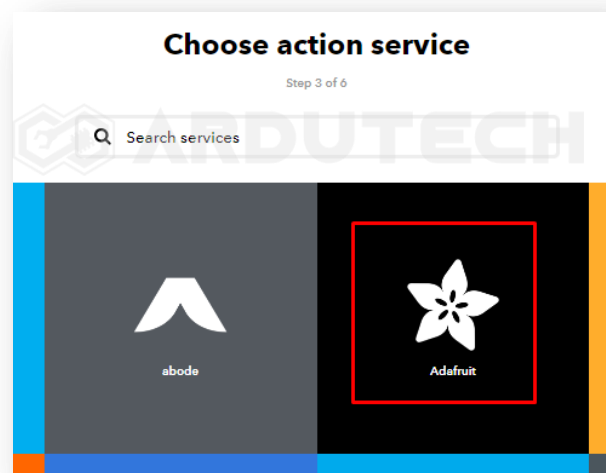
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik **"Create trigger"**.

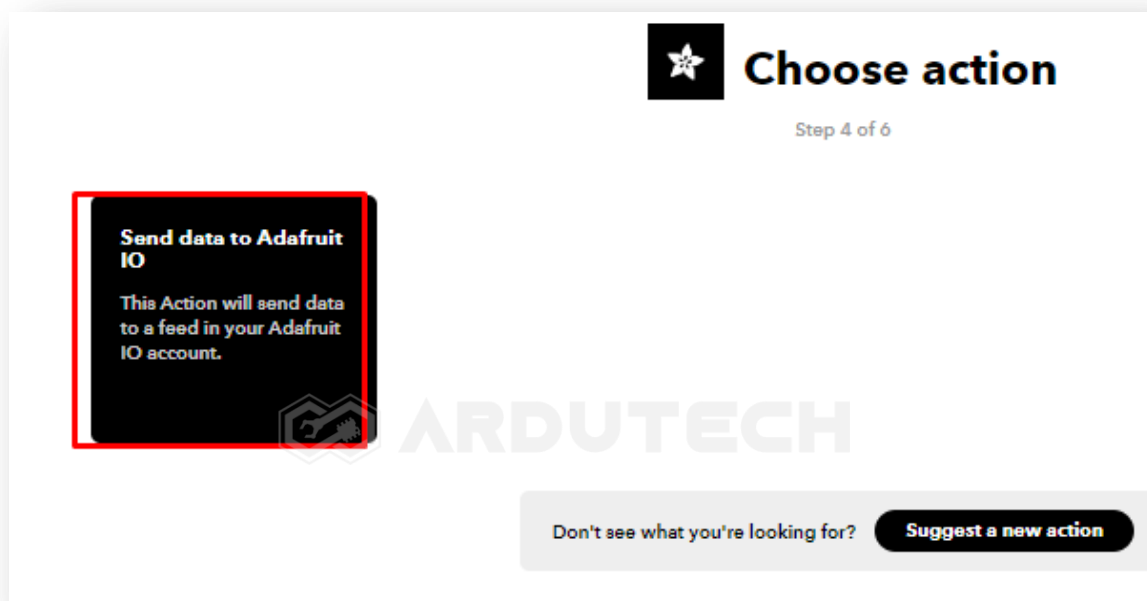
Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If Then That

Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields

Step 5 of 6

Feed name

Relay3

The name of the feed to save data to.

Data to save

1

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay3**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 3 On",
then Send data to
Relay3 feed

54/140

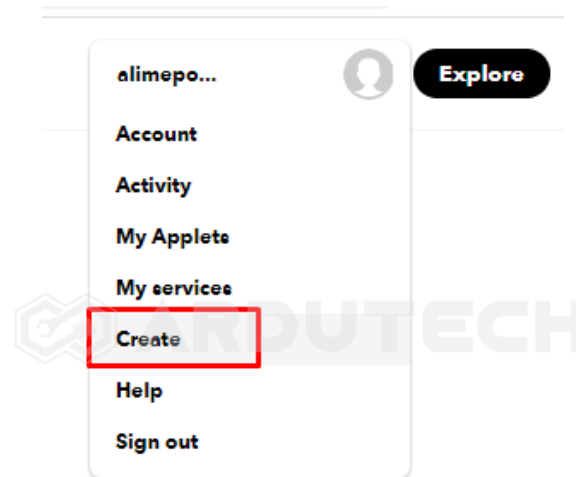
by alimepol22

Receive notifications
when this Applet runs

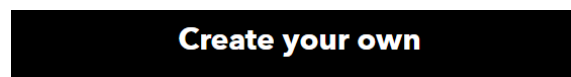
Finish

Terakhir klik **Finish**.

Pilih lagi menu **"Create"** :

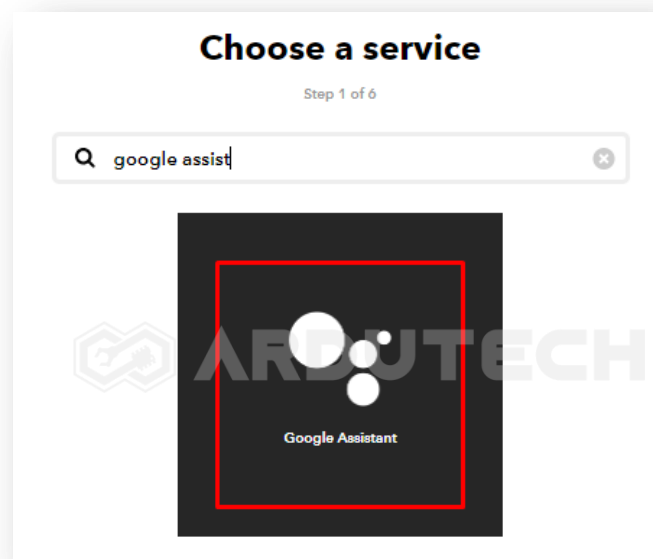


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.



If **+** This Then That

Klik + **This** sehingga tampil :



Pada kolom pencarian ketik “**google assistant**”, setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".



Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 3 off

What's another way to say it? (optional)

Relay three off

And another way? (optional)

Three off

What do you want the Assistant to say in response?

Ok boss

Language

English



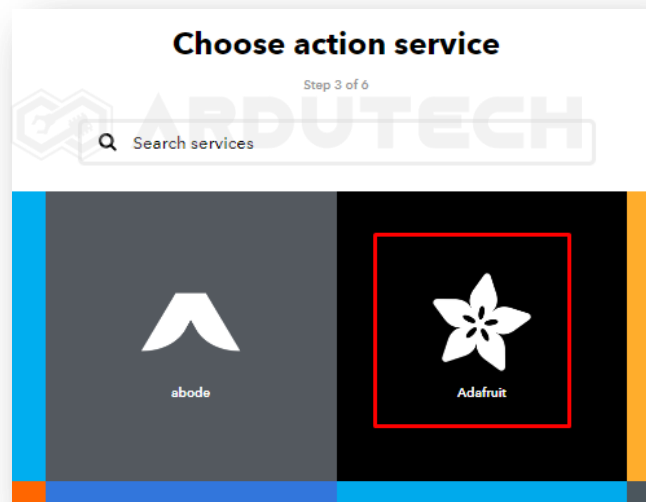
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik "Create trigger".

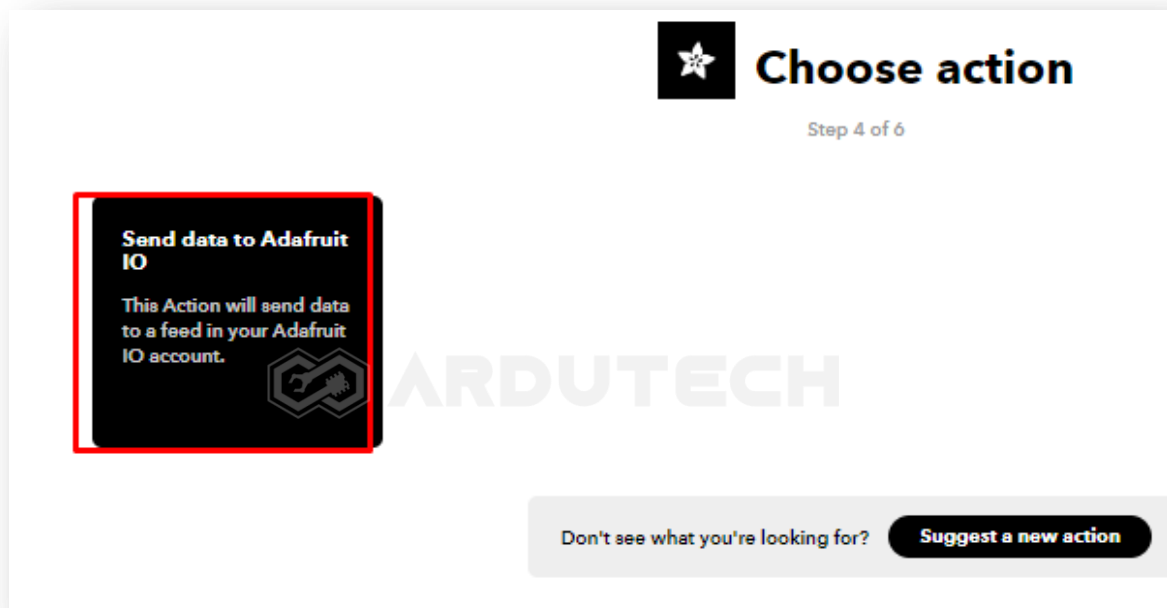
Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If Then That

Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields

Step 5 of 6

Feed name

Relay3

The name of the feed to save data to.

Data to save

0

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay3**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 3 off", then Send data to Relay3 feed

55/140

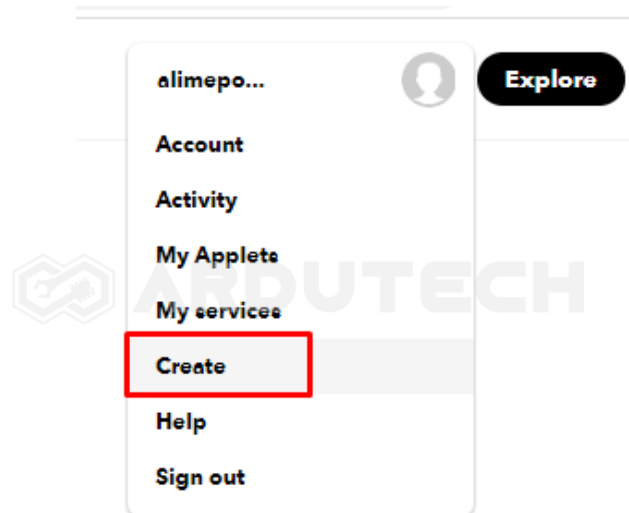
by alimepol22

Receive notifications when this Applet runs

Finish

Terakhir klik **Finish**.

Pilih lagi menu "**Create**" :

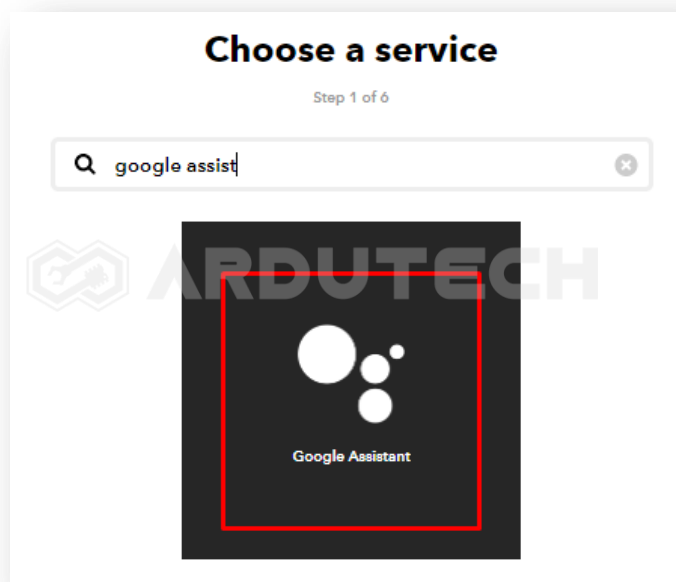


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.

Create your own

If **+** This Then That

Klik + This sehingga tampil :



Pada kolom pencarian ketik “**google assistant**”, setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

< Back



Choose trigger

Step 2 of 6

Say a simple phrase

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient

This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for?

Suggest a new trigger

Pilih "Say a simple phrase".



Complete trigger fields

Step 2 of 6

What do you want to say?

Relay 4 On

What's another way to say it?
(optional)

Relay four On

And another way? (optional)

Four On

What do you want the Assistant to say
in response?

Ok boss

Language

English



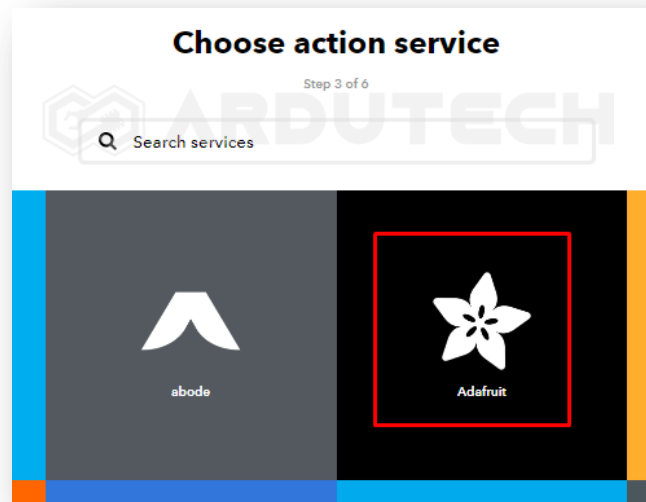
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik "Create trigger".

Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If Then That

Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields

Step 5 of 6

Feed name

Relay4

The name of the feed to save data to.

Data to save

1

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay4**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 4 On", then Send data to Relay4 feed

54/140

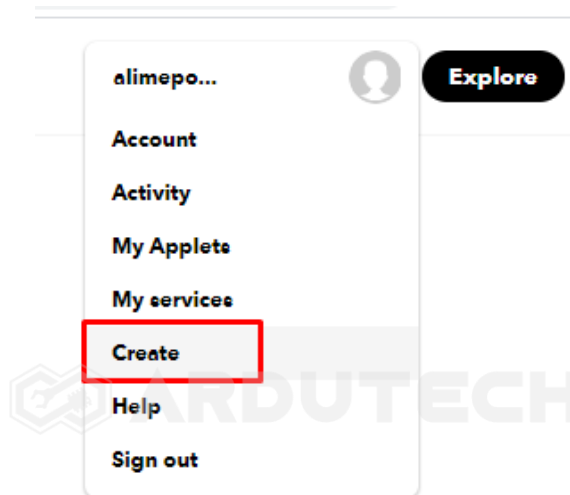
by alimepol22

Receive notifications when this Applet runs

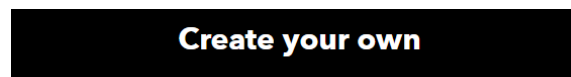
Finish

Terakhir klik **Finish**.

Pilih lagi menu "**Create**" :

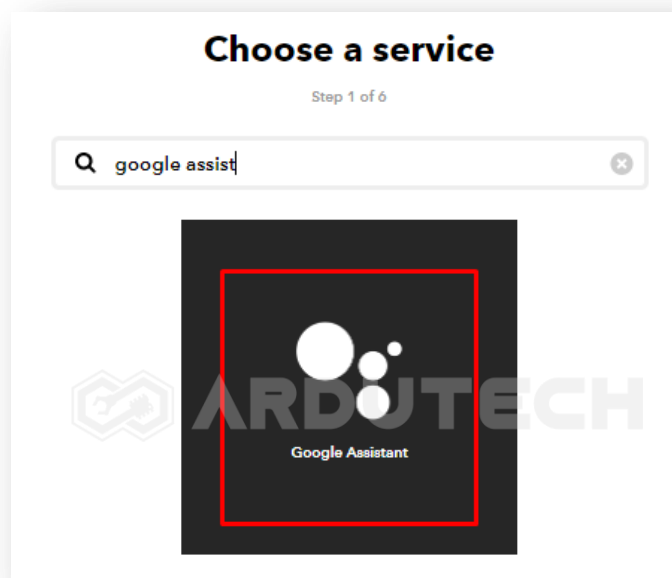


Tampilan dan proses sama dengan membuat Applet untuk Relay 1.



If **+** This Then That

Klik **+** This sehingga tampil :



Pada kolom pencarian ketik **“google assistant”** , setelah muncul Google Assistant kemudian diklik maka akan tampil halaman untuk memilih trigger.

IFTTT Home Search

[Back](#)

 **Choose trigger**

Step 2 of 6

Say a simple phrase
This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase you choose. For example, say "Ok Google, I'm running late" to text a family member that you're on your way home.

Say a phrase with a number
This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Set Nest thermostat to 68." **Use the # symbol to specify where you'll say the number ingredient

Say a phrase with a text ingredient
This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Post a tweet saying 'New high score.'" **Use the \$ symbol to specify where you'll say the text ingredient

Say a phrase with both a number and a text ingredient
This trigger fires when you say "Ok Google" to the Google Assistant followed by a phrase like "Block time for 'exercise' at 6 PM." **Use the # symbol to specify where you'll say the number ingredient and \$ where you'll say the text ingredient

Don't see what you're looking for? [Suggest a new trigger](#)

Pilih "Say a simple phrase".

 **Complete trigger fields**

Step 2 of 6

What do you want to say?

Relay 4 Off

What's another way to say it? (optional)

Relay four Off

And another way? (optional)

Four off

What do you want the Assistant to say in response?

Ok boss

Language

English

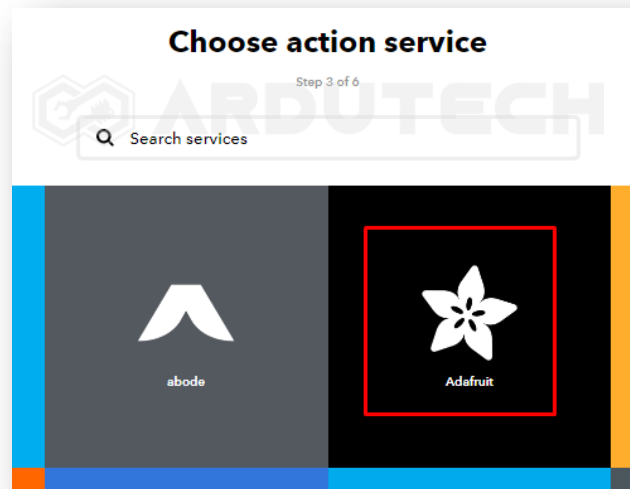
Create trigger

Muncul kolom untuk membuat phrasa/kata 'perintah kontrol' ke google assistant. Isi seperti pada gambar diatas atau anda dapat membuatnya sendiri. Selesai klik "**Create trigger**".

Nah tadu itu kita sudah membuat bagian **IF** nya, sekarang kita masuk ke bagian **THAT**. Klik **+ THAT**.

If Then That

Cari aplikasi /service dari **Adafruit**.



Klik Adafruit :



Klik bagian sebelah kiri : **Send data to Adafruit IO**.



Complete action fields

Step 5 of 6

Feed name

Relay4

The name of the feed to save data to.

Data to save

0

The data to be saved to your feed.

Add ingredient

Create action

Bagian ini ada kaitannya dengan Adafruit, pilih **Feed name** tadi yang sudah dibuat di Adafruit yaitu **Relay4**. Kemudian bagian **Data to save** merupakan 'sinyal kontrol' nya kita isi 0, maksudnya logika '0' untuk 'OFF'.

Klik **Create action**.

Review and finish

Step 6 of 6

If You say "Relay 4 Off",
then Send data to
Relay4 feed

55/140

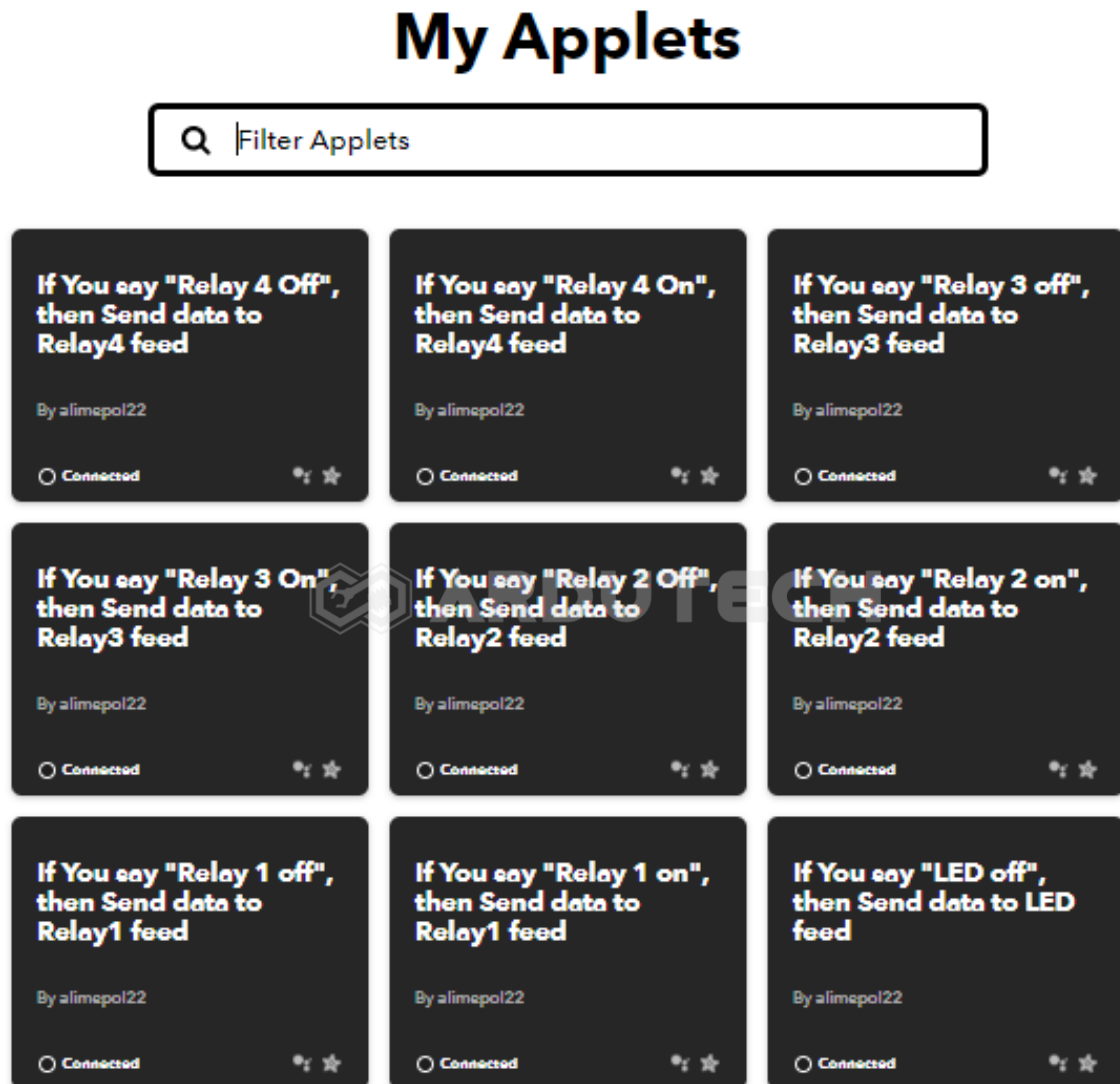
by alimepol22

Receive notifications
when this Applet runs

Finish

Terakhir klik **Finish**.

Ok sekarang kita sudah punya 8 applet untuk mengontrol 4 Relay :



Program/Source Code di Arduino IDE

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h
- Adafruit_MQTT.h
- Adafruit_MQTT_Client.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/*  
*****  
* Project Kontrol 4 Relay dengan Google Assistant  
* Board : NodeMCU ESP8266 V3  
* Input : Google Assistant  
* Output : 4 Relay  
* 99 Proyek IoT  
*/
```

* www.ardutech.com

*****/

```
#include <ESP8266WiFi.h>
```

```
#include "Adafruit_MQTT.h"
```

```
#include "Adafruit_MQTT_Client.h"
```

```
#define RL1      D5
```

```
#define RL2      D6
```

```
#define RL3      D7
```

```
#define RL4      D8
```

//---GANTI SESUAI DENGAN JARINGAN WIFI

//---HOTSPOT ANDA

```
#define WLAN_SSID  "ArdutechWiFi" // Nama Hotspot/WiFi
```

```
#define WLAN_PASS  "12345678" // Password
```

```
#define AIO_SERVER  "io.adafruit.com"
```

```
#define AIO_SERVERPORT 1883
```

//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA

```
#define AIO_USERNAME  "ArdutechJogja"
```

//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA

```
#define AIO_KEY      "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"
```

```
WiFiClient client;
```

```
Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT, AIO_USERNAME, AIO_KEY);
```

```
Adafruit_MQTT_Subscribe Relay1 = Adafruit_MQTT_Subscribe(&mqtt, AIO_USERNAME"/feeds/Relay1"); // FeedName
```

```
Adafruit_MQTT_Subscribe Relay2 = Adafruit_MQTT_Subscribe(&mqtt, AIO_USERNAME"/feeds/Relay2");
```

```
Adafruit_MQTT_Subscribe Relay3 = Adafruit_MQTT_Subscribe(&mqtt, AIO_USERNAME"/feeds/Relay3");
```

```
Adafruit_MQTT_Subscribe Relay4 = Adafruit_MQTT_Subscribe(&mqtt, AIO_USERNAME"/feeds/Relay4");
```

```
void MQTT_connect();  
//=====
```

```
void setup() {  
  Serial.begin(9600);  
  pinMode(RL1, OUTPUT);  
  pinMode(RL2, OUTPUT);  
  pinMode(RL3, OUTPUT);  
  pinMode(RL4, OUTPUT);  
  
  // Connect to WiFi access point.  
  Serial.println(); Serial.println();  
  Serial.print("Connecting to ");  
  Serial.println(WLAN_SSID);  
  
  WiFi.begin(WLAN_SSID, WLAN_PASS);  
  while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print(".");  
  }  
  Serial.println();  
  
  Serial.println("WiFi connected");  
  Serial.println("IP address: ");  
  Serial.println(WiFi.localIP());  
  mqtt.subscribe(&Relay1);  
  mqtt.subscribe(&Relay3);  
  mqtt.subscribe(&Relay2);  
  mqtt.subscribe(&Relay4);  
}
```

```
//=====
```

```
void loop() {  
  MQTT_connect();  
  Adafruit_MQTT_Subscribe *subscription;  
  while ((subscription = mqtt.readSubscription(20000))) {  
    if (subscription == &Relay1) {  
      Serial.print(F("Got: "));  
      Serial.println((char *)Relay1.lastread);  
      int Relay1_State = atoi((char *)Relay1.lastread);  
      digitalWrite(RL1, Relay1_State);  
    }  
    if (subscription == &Relay2) {  
      Serial.print(F("Got: "));  
      Serial.println((char *)Relay2.lastread);  
      int Relay2_State = atoi((char *)Relay2.lastread);  
      digitalWrite(RL2, Relay2_State);  
    }  
    if (subscription == &Relay3) {  
      Serial.print(F("Got: "));  
      Serial.println((char *)Relay3.lastread);  
      int Relay3_State = atoi((char *)Relay3.lastread);  
      digitalWrite(RL3, Relay3_State);  
    }  
    if (subscription == &Relay4) {  
      Serial.print(F("Got: "));  
      Serial.println((char *)Relay4.lastread);  
      int Relay4_State = atoi((char *)Relay4.lastread);  
      digitalWrite(RL4, Relay4_State);  
    }  
  }  
}
```

```
}  
//=====  
void MQTT_connect() {  
    int8_t ret;  
    if (mqtt.connected()) {  
        return;  
    }  
    Serial.print("Connecting to MQTT... ");  
    uint8_t retries = 3;  
    while ((ret = mqtt.connect()) != 0) { // connect will return 0 for connected  
        Serial.println(mqtt.connectErrorString(ret));  
        Serial.println("Retrying MQTT connection in 5 seconds...");  
        mqtt.disconnect();  
        delay(5000); // wait 5 seconds  
        retries--;  
        if (retries == 0) {  
            // basically die and wait for WDT to reset me  
            while (1);  
        }  
    }  
    Serial.println("MQTT Connected!");  
}
```

PERHATIKAN !

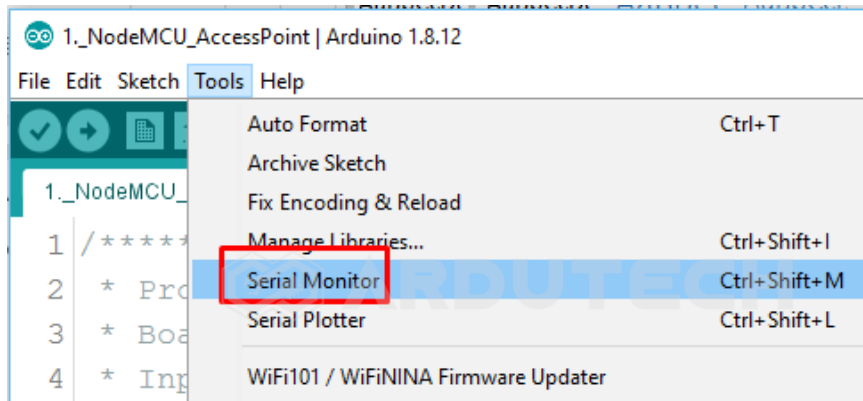
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **WLAN_SSID**
- Password jaringan WiFi/hotspot : **WLAN_PASS**
- Username Adafruit IO : **AIO_USERNAME**
- Active Key Adafruit IO : **AIO_KEY**

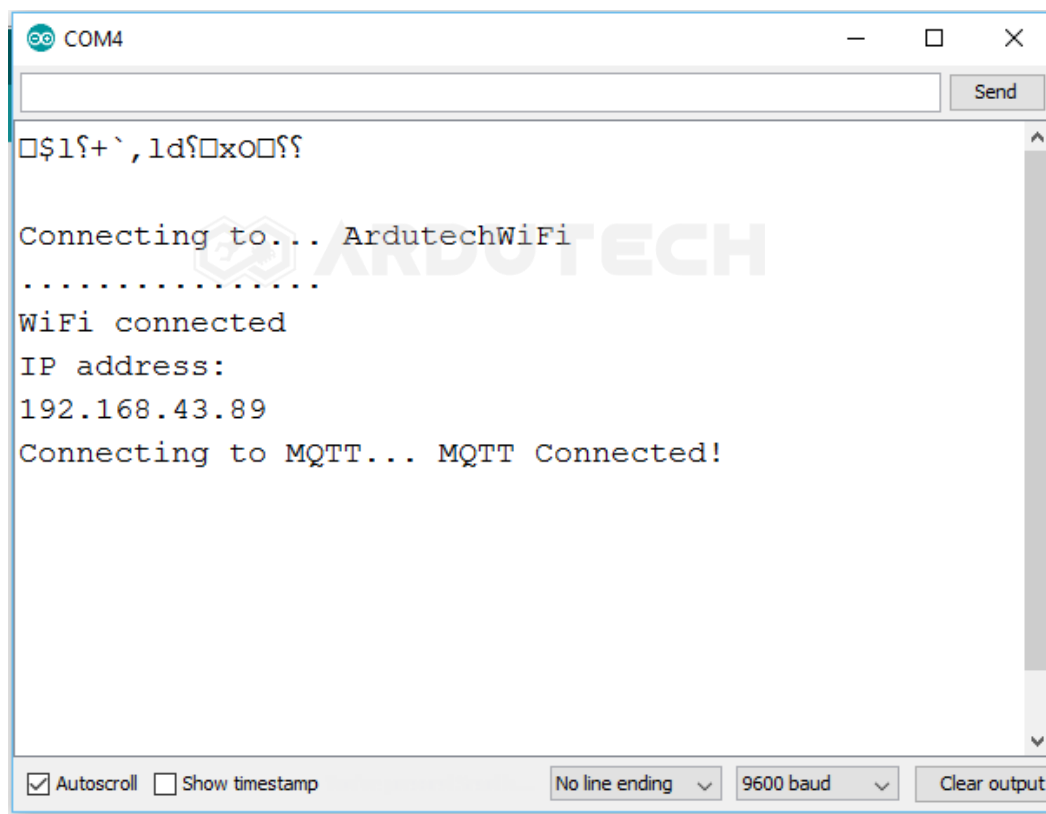
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

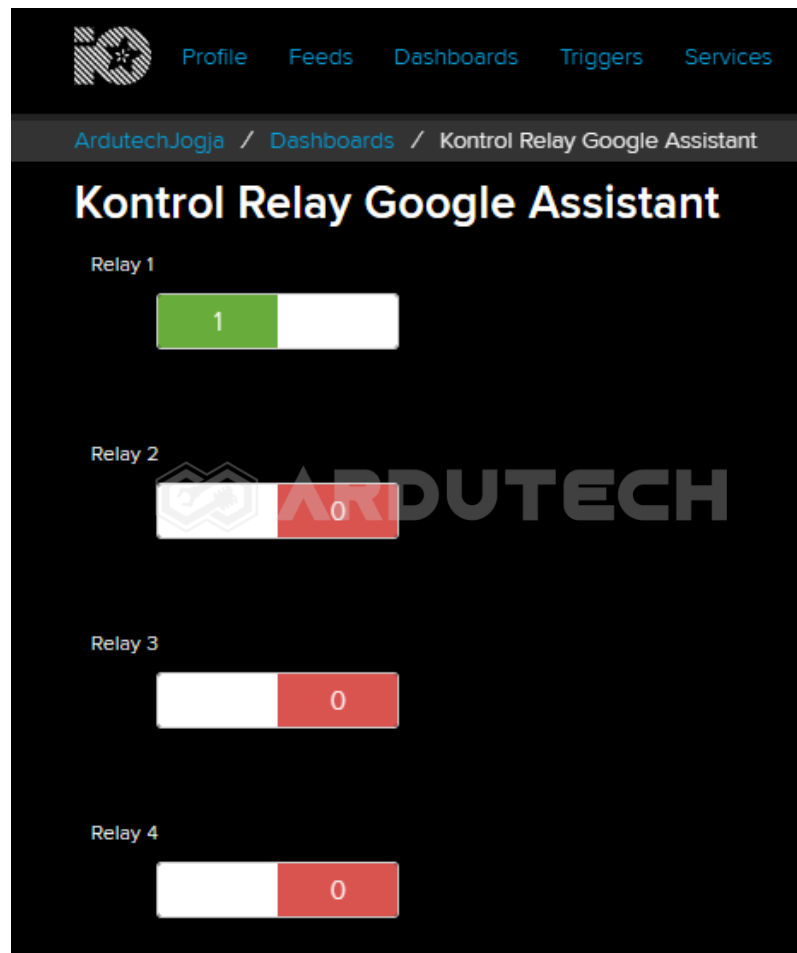
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools** → **Serial Monitor** kemudian seting baudrate pada nilai 9600.



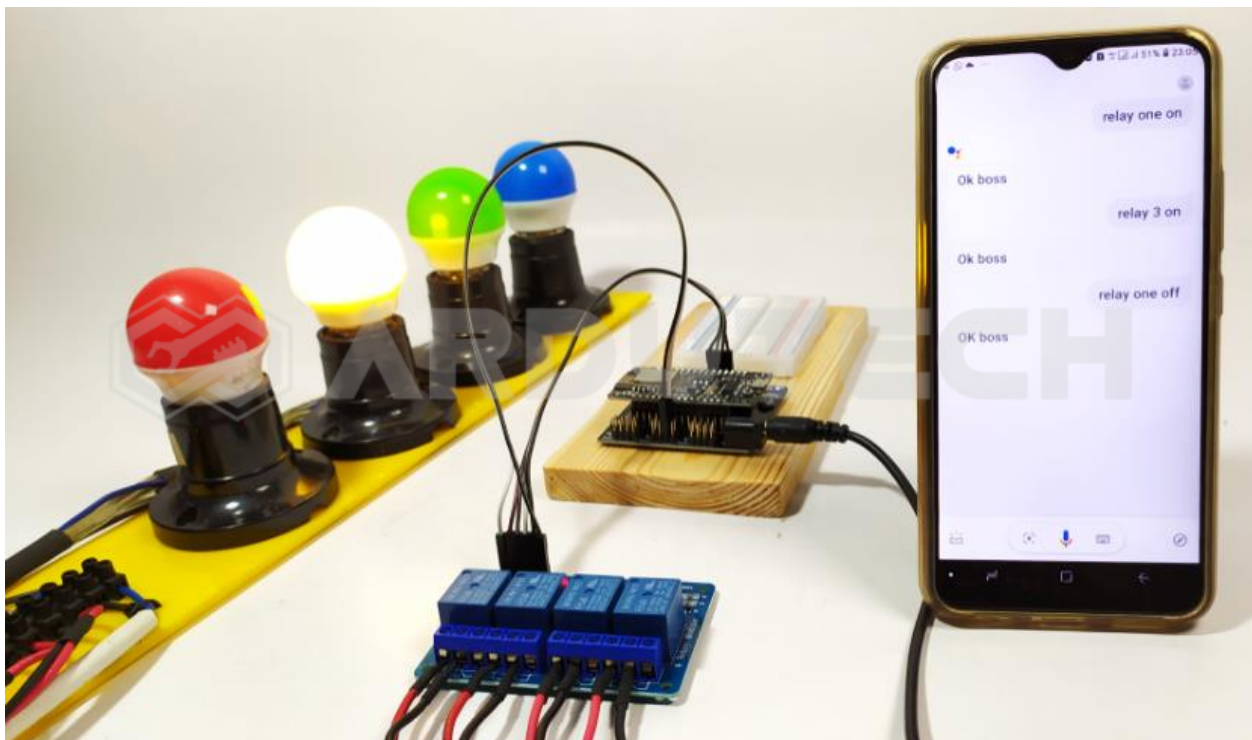
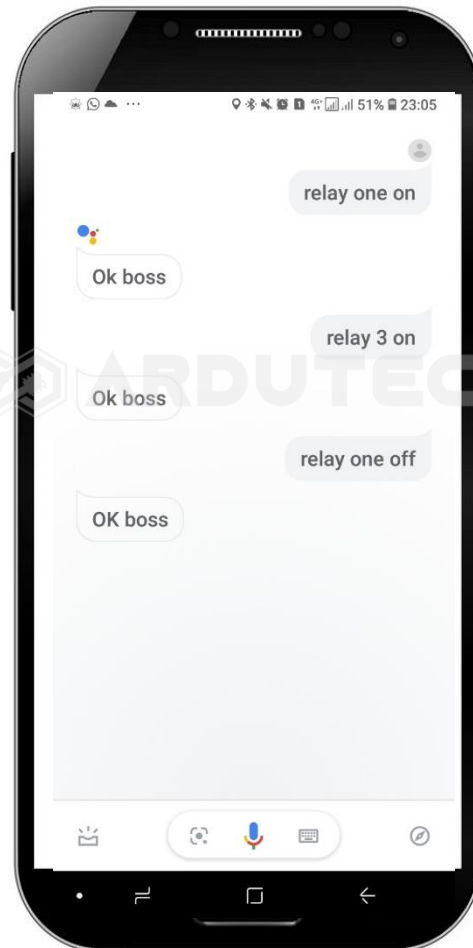
Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Tekan tombol ON – OFF pada block Relay 1 dan perhatikan Relay 1 yang terhubung dengan pin D5 NodeMCU.



Jika berhasil maka Relay 1 akan nyala "ON" ketika tombol ditekan "ON". Nah sekarang kita kontrol dengan Google Assistant. Install aplikasi Google Assistant. Ucapkan perintah dengan jelas "relay one on" dengan bahasa Inggris, jika command tidak sesuai maka perintah tidak dikenali, jika sesuai maka akan ada respon dari google kemudian Relay 1 juga akan ON.



Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – “Sahabat Inovasi Anda”